ERIC GABRIEL MANABAT

FULL-STACK AI ENGINEER

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EDUCATION

Pamantasan ng Lungsod ng Valenzuela | *Bachelor of Science in Information Technology* 2023 - 2027
Honors: **Consistent Dean's Lister (1.03 Running GWA)** | Valenzuela City, Philippines

PROFESSIONAL EXPERIENCE

FlyRank AI | *Full-Stack AI Engineer Intern* Jul 2026 - Sep 2026 | Remote

- Architected and deployed an autonomous FastAPI AI sidecar microservice, increasing retrieval accuracy from **81.8% to 86.4% Top-1** on ground-truth benchmarks by engineering hybrid BM25 + dense vector search with Reciprocal Rank Fusion (RRF k=60) and deterministic blend reranking.
- Eliminated out-of-domain hallucinations to achieve a **100% negative query pass rate** and **90% LLM-as-judge relevance score** by designing a bounded 3-step agentic query loop with function calling and tool use, cosine similarity gating (<0.50 threshold), and inline citation grounding.
- Reduced time-to-first-token (TTFT) by **~65%** and established real-time operational observability by implementing asynchronous Server-Sent Events (SSE) token streaming APIs, Prometheus /metrics latency histograms, and Grafana telemetry dashboards.
- Delivered full-stack streaming chat interfaces validated across **78 pytest integration tests** and a **27-case golden-set evaluation suite**, eliminating regression failures across synthetic edge cases.

Nexvision Innovations Inc. | *Full Stack Engineer (Lead Intern)* Jul 2026 - Sep 2026 | Marikina (Hybrid)

- Engineered DOLE statutory compliance algorithms across 4 enterprise applications by designing interval-partitioned time calculations to isolate 10 PM - 6 AM night differentials from standard overtime multipliers.
- Automated statutory deduction and 13th-month proration calculations across **500+ employee records** by developing modular TypeScript/Next.js computation engines with immutable audit logging.
- Hardened multi-tenant PostgreSQL/Supabase data layers against cross-branch data leaks via row-level security (RLS), and cut monthly audit prep time from **3 days to under 5 minutes** via high-throughput Excel export engines.
- Accelerated CI/CD test execution by **40% (~12m to ~7m)** via an offline mock testing daemon in GitHub Actions and Jenkins, and led sprint execution across 180+ Jira/GitHub issues as Intern Team Lead.

FEATURED AI PROJECTS

PLV eLib - Academic Library AI Sidecar | *FastAPI, Laravel, PostgreSQL, SQLite FTS5, Prometheus, Grafana* 2026

- Architected decoupled dual-engine topology pairing a Laravel 11 monolith with an external FastAPI AI sidecar, lifting Top-1 retrieval accuracy from **81.8% to 86.4%** via SQLite FTS5 (BM25) and dense vectors fused with RRF (k=60).
- Constructed bounded agentic retrieval loop with function calling and tool use, deterministic refusals on out-of-domain queries, and automated LLM-as-judge verification, achieving a **100% negative query pass rate**.
- Deployed Prometheus metrics and a 6-chart Dockerized Grafana dashboard to track production inference latency histograms and live query feedback streams.

AGOS - AI-Guided Overflow Surveillance | *FastAPI, React, YOLOv8, Groq, Docker, Raspberry Pi* 2026

- Fine-tuned custom YOLOv8 object detection model on **5,000 CVAT-annotated** drainage images, automating real-time floating debris and waterway blockage classification.
- Engineered real-time Groq LPU telemetry synthesis service streaming structured flood risk assessments via SSE, integrating multi-model failover that guaranteed zero-downtime analytics across live sensor logs.
- Slashed cloud GPU compute overhead by **92%** by architecting a decoupled batch inference engine that evaluated camera streams on an optimized 3-minute inference cadence.
- Formulated multi-modal hazard scoring engine fusing visual blockage, ultrasonic depth, and weather APIs to trigger 3-tier physical LEDs and queued emergency push/SMS alerts with a 30-minute deduplication cooldown.

TECHNICAL SKILLS

AI & Machine Learning: Function Calling & Tool Use, RAG Pipelines, LangChain, Vector Search (pgvector, Pinecone, ChromaDB, SQLite FTS5 (BM25), Hybrid RRF), Embeddings, YOLOv8, Groq, OpenCV, LLM-as-Judge Evaluation, Prompt Engineering, SSE Streaming, Prometheus & Grafana Observability

Languages: Python, TypeScript, JavaScript, SQL, PHP, C# / .NET, Java

Frameworks & Full-Stack: FastAPI, Next.js (App Router), React 19, Node.js, Laravel 11/12, Tailwind CSS v4, Livewire 3

Databases & Cloud: PostgreSQL, Supabase, Prisma ORM, Drizzle ORM, Redis, Docker, AWS, Google Cloud (GCP), GitHub Actions, Jenkins

Cybersecurity & Tools: Burp Suite, Wireshark, Kali Linux, Role-Based Access Control (RBAC), Row-Level Security (RLS)

HONORS & CERTIFICATIONS

- **LLM Zoomcamp - Production LLM & RAG Engineering Bootcamp** | DataTalks.Club | Aug 2026
- **Learn RAG (Retrieval-Augmented Generation)** | Boot.dev | 2026
- **Global Cyber Skills Benchmark (Project Nightfall)** - Ranked 63/589 Global Teams | HackTheBox | May 2026
- **Google Cybersecurity Professional Certificate** | Google | Jan 2026
- **Mastering REST APIs with FastAPI** | Packt / Coursera | May 2026
- **ITLympics 2026 - Silver Medal** | Capture the Flag | Pamantasan ng Lungsod ng Valenzuela | Mar 2026